Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/06/2015
PI Name:	Matin, A.C. Ph.D.		
Project Title:	Spaceflight Effects on Bacterial Antibiotic Resistance and Its Genetic Basis “AntimicrobialSat”		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	A.Matin@stanford.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(650) 725-4745
Organization Name:	Stanford University		
PI Address 1:	299 Campus Drive, West		
PI Address 2:	Mail Code 5124		
PI Web Page:			
City:	Stanford	**State:**	CA
Zip Code:	94305-5101	**Congressional District:**	14
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 SALMON--Astrobiology & FSB NNH08ZDA009O-SCMAFSB
Start Date:	07/01/2010	**End Date:**	03/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	Small Satellites		
Flight Assignment:	NOTE: End date changed to 3/31/2016 per NSSC information (Ed., 1/20/16) NOTE: End date changed to 11/30/2015 per NSSC information and A. Chu/ARC (Ed., 12/16/14) NOTE: New end date is 11/30/2014 per NSSC information (Ed., 4/4/14) NOTE: New end date is 4/30/2014 per A. Chu/ARC and NSSC (Ed., 2/21/2013)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX10AM90A		
Performance Goal No.:			
Performance Goal Text:			

This project is concerned with the design, development, and testing of a nanosatellite payload, and the proposed biological experiment after launch in 2013. As in the past, the NASA and Stanford teams have worked closely in this project period to optimize the experimental conditions and ensure the success of the experiment in space. This includes weekly conference calls, visits by the parties to each other's institutions, and regular discussion in more formal setting in the presence of other NASA staff. The two component personnel have enjoyed a close, pleasant, and productive interactive relationship.
As dramatized in the movie Apollo 13, astronauts during spaceflight can contract bacterial infections. Astronauts in space have also experienced infections with other opportunistic and pathogenic microbes. Effective treatment of bacterial infections has required therapy customized for the space environment, as evidence has accumulated that microbes respond differently to antimicrobials in the space environment (1). The EcAMSat experiment focuses on directly documenting alterations in antibiotic resistance in the space environment using a well-defined microbial system - the bacterium, E. coli, and a widely-used antibiotic, gentamicin, against a uropathogenic strain of this bacterium.

Integrated bioreactors have flown on manned spacecraft; they require humans to initiate experiments, record data, and collect or prepare samples to return them to Earth for characterization. Fully autonomous microsystems like EcAMSat reduce the cost and obviate other limitations of human-tended space biological experiments, while expanding significantly the number of launch opportunities.

The Matin group has shown in Earth experiments that bacterial antibiotic resistance depends on their intact innate resistance, whose central regulator is the stress sigma factor, Sigma S (ss). Further, our studies with simulated microgravity obtained using high aspect ratio (HARV) bioreactors, have shown that in stationary phase, the bacterial antibiotic resistance is greatly enhanced, again dependent on this sigma factor (encoded by the rpoS gene) (3). Our recent findings have identified several proteins controlled by this sigma factor that determine E. coli resistance to gentamicin (4). Future work to assess the role of these genes in space can be expected to provide further novel means of countering bacterial resistance in space.

EcAMSat's 14 kg mass comprises a structure roughly 10 x 22 x 34 cm, including triple-junction solar cells; patch antenna for 2.4-GHz bidirectional communication using a commercial WiFi-technology radio; self-deploying omnidirectional "beacon" antenna for spacecraft location and education-and-outreach broadcasting on the amateur ("ham") band (437.465 MHz); mechanical structure; spacecraft power/processor/communications/control system (collectively, the "bus"); and the experiment containment vessel, internal volume 1.2 L, which houses the bio/fluidic, opti-cal, thermal, sensor, and payload electronics subsystems. Magnetic rods passively orient the satellite, aiming the spacecraft's long axis and hence the patch antenna towards Earth's north magnetic pole during the part of each ~93-minute orbital period when the spacecraft is above the northern hemisphere. Magnetic hysteresis rods damp nutation or "wobble" in spacecraft motion by dissipating energy as heat when they move across Earth's magnetic field lines.

The EcAMSat fluidic system includes forty-eight 100-µL culture wells, each 4 mm in diameter x 7.8 mm deep. Use of 9-mm pitch along the rows and 9- and 13.5-mm center-to-center row spacing allows fluidic card optical analysis with standard multiwell plate readers. The fluidic card was fabricated by Micronics, Inc. from laser-cut poly(methylmethacrylate) layers laminated to one another using pressure-sensitive acrylic adhesive interlayers supported on PET (poly(ethylene terephthalate)) carrier films. Four independent sets of manifolded fluidic inlets and outlets supply nutrients and antibiotic agent to the four 12-well banks. Bacterial cells are confined in the microwells by integral 0.2-µm pore-size nylon fiber membrane filters (Sterlitech). The NASA group with Stanford participation and input has continued to develop and test the nanosatellite payload, including the biological experiment that will test the hypothesis that stress response to the spaceflight microgravity environment enhances antibiotic resistance in uropathogenic E. coli (UPEC) in a manner dependent on its stress transcriptional factor, ss. Based on the results, a protocol has been developed for the space experiments.

The pressure values for introducing various fluids into microwells is nearly optimized and the full protocol will begin to be tested on Earth in the next few weeks. Also, a standard curve for survival of starving experimental bacteria is being determined to fend for unforeseen alteration in the stowage period before the commencement of the flight. Similarly, preparations are being made for parallel control experiments on Earth during the spaceflight.

EcAMSat's "2U" payload will support wild-type UPEC and its isogenic rpoS mutant (lacking sS, which is encoded by the rpoS gene) in four independent banks of 12 100-µL wells (6 wells each for wild-type and mutant) in a microfluidic card that delivers nutrient, confines cells using integral 0.2-µm-pore filters, and maintains temperature (37°C). Each microwell (pathlength: 8 mm) has a dedicated 3-color LED and photodetector to track growth of the organisms via light scattering (470 nm). Cellular metabolism is tracked using Alamar Blue indicator dye (615 nm) and its reduced form (525 nm), generated by bacterial metabolism.

Task Description:

Growth and metabolism of E. coli are measured versus the concentration of the antibiotic gentamicin (four well banks provide three concentrations and a control). Following orbital stabilization, bacteria in stasis are given nutrient and grown to stationary phase, the phase most important in disease causation. The bacteria are then rinsed with nutrient-free buffer and starved for 24 hr (to firm up the innate resistance of stationary phase cells). Finally, they are treated with gentamicin for 24 hours, followed by tracking survival dynamically via Alamar Blue. We have performed detailed baseline and feasibility experiments, showing that Alamar Blue measurements reliably indicate the increased sensitivity of the mutant to gentamicin.

As stated above, the rationale of the project is based on the fact that there is convincing evidence that microgravity (MG) increases bacterial resistance and weakens the human immune response, which necessitates careful examination of bacterial threat to astronauts in spaceflight. We previously discovered, using simulated microgravity [called low shear simulated microgravity (LSMMG)], using the high aspect to volume (HARV) bioreactors, that the resistance of uropathogenic E. coli (UPEC) to the antibiotic gentamicin (Gm) increases under these conditions, and this increase is dependent on a gene called rpoS. This bacterium causes cystitis, incidence of which has been reported in astronauts; Gm is used to treat this disease.

NNX10AM90A ("EcAMSat") is aimed at testing, under actual MG of spaceflight, the validity of the above findings. This is a collaborative effort between Stanford and NASA Ames Research Center at Moffett Field. If confirmed in space experiments, the results would indicate effective countermeasures to enhance the safety of space travel.

Our teams –at Stanford and at Moffett Field - have achieved notable success. First, we have established a completely reliable system for monitoring changes in UPEC activity by a color reaction that can be used to beam real time data from space (in this case from a nanosatellite) via telemetry. These experiments were done at Stanford using microtiter plate

wells. Cells were grown to stationary phase, starved (which is necessary to activate the stress response, and is a condition bacteria often experience in disease causation process), then diluted to an A660 of 0.45 in M9 salts (a kind of buffer) with the addition of 16 µg/mL gentamicin (Gm), incubated at 37C for 24 hours, after which they were dispensed in microtiter plate wells and Alamar Blue (AB) was added to interrogate the system for viability – the viability in this system is shown by the extent of the conversion of AB from blue to pink color. Controls without the drug etc. were included. The results clearly showed that Gm addition has much less lethal effect on the wild type than on the mutant; this was evident both from the remaining blue color in the plate wells, and the corresponding increase in red color, as well as the graphic representation of the color changes. Thus the loss of the gene that is missing from the mutant (rpoS) makes it more sensitive to the drug, and this can be documented by the differing loss of the capacity to convert AB from blue to pink).

A lot of work was needed to adapt the payload apparatus to work with bacteria, the beginning point being the PharmaSat hardware that was used for space experiments with Yeast. Filters needed to be changed, biocompatibility, cell escape issues, fluidics timing, and pumping to accommodate the new fluid flow characteristics, all needed to be addressed and now meet the engineering requirements. A number of tests including certain experimental verification tests (EVTs) have been run, and it is confirmed in preliminary experiments that the hardware can track cell activity by permitting monitoring of AB color change.

The outcome is that the system as a whole has now begun to work and is now ready to be optimized. One major issue is to determine the amount of pressure that can be used during fluid exchanges (such as exchanging the growth medium with the antibiotic) without causing cells to be swept out in large numbers from the suspension on to the filters. This and related aspects are now being addressed. We are confident that these efforts will shortly provide a new hardware system that would be generically capable of experiments with bacteria in spaceflights.

References:

1. Principles of Clinical Medicine for Space Flight M. R. Barratt, S. L. Pool, Eds. (Springer, New York, ed. 1, 2008).

2. Ricco, A.J., Hines, J.W., Piccini, M., Parra, M., Timucin, L., Barker, V., Storment, C., Friedericks, C., Agasid, E., Beasley, C., Giovangrandi, L., Henschke, M., Kitts, C., Levine, L., Luzzi, E., Ly, D., Mas, I., McIntyre, M., Oswell, D., Rasay, R., Ricks, R., Ronzano, K., Squires, D., Swaiss, G., Tucker, J., and Yost, B., “Autonomous genetic analysis system to study space effects on microorganisms: Results from orbit,” Proc. Transducers 2007, IEEE: New York (2007); pp. 33-37.

3. Singh, R, and Matin, A C. unpublished

4. Submitted, PLoS Biology

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Antibiotic resistance in bacteria is becoming an extremely serious problem on Earth that will impact space travellers as well (the astronauts carry with them the Earth microflora). The research underway will provide a critical test for a hypothesis, which if confirmed, will show the way to greatly mitigate bacterial resistance to antibiotics in general. This hypothesis is based on a firm foundation provided by experiments conducted on Earth; and since bacterial resistance increases under microgravity conditions, its testing in space provides an amplified situation where the difference made by the absence of the gene responsible can be seen more clearly.
The innovative technology that is being developed is a generation of small molecules that can silence the gene in question, which can then either be used alone (since the resulting weakened bacteria can be more easily dealt with by the immune system) or in combination with existing antibiotics for effective treatment of infectious disease here on Earth as well as in space.

As explained in previous reports the EcAMSat project is based on the hypothesis that UPEC resistance to the antibiotic gentamicin (Gm) will become greater under space microgravity, and its dependence on the stress sigma factor, Sigma S, will become more marked under these conditions. The importance of exploring this phenomenon lies both for space travel as well as for life on Earth. Bacterial antibiotic resistance has become a very serious overall public health threat and this development is particularly hazardous for astronauts for two reasons: their immune system is weakened and bacteria become more resistant to disinfectant agents, including antibiotics, under microgravity conditions.
The Sigma S hypothesis is supported by our finding in low-shear modeled microgravity. We have recently published an extensive analysis of the mechanism involved (see the link: http://web.stanford.edu/).

To summarize the critical finding: bacteria protect themselves against antibiotics of different classes using their antioxidant defense machinery. The current efforts to overcome bacterial resistance are antibiotic-specific; thus for instance, a measure taken to overcome bacterial resistance to ß-lactams is to neutralize ß-lactamase that resistant bacteria produce. Our discovery of the involvement of antioxidant defense in bacterial antibiotic resistance means that an overall way to enhance the efficacy of antibiotics against bacteria is to compromise their antioxidant defense machinery. This can be done by discovering small molecule compounds that can inhibit the proteins of this defense that we have implicated in bacterial antibiotic resistance.

In order to be able to determine if the Sigma S hypothesis is valid under the conditions of actual microgravity, this project will perform experiments aboard a small satellite using the Gm. A way has to be found of determining the effect of Gm on bacterial activity in a way that could be beamed from space to Earth via telemetry. We have perfected this method at Stanford by using Alamar Blue to measure bacterial metabolic activity. The speed of color change from blue to magenta of this dye indicates bacterial viability, and we successfully showed that this color change accurately duplicates the results of plate counts in showing that the wild type is more resistant to Gm than its isogenic mutant missing Sigma S. More recently, we have been attempting to show that this can be done in the space hardware as well. Recent results have given promise, and more tests are in progress; we believe that the experimental system would be ready to fly very shortly.

Several publications and abstracts have so far resulted, as shown below. We have also written a book chapter, but while

Task Progress:	our chapter has been finalized, the book has not yet gone to the publisher: 1. J-H Wang, R Singh, M Benoit, M Keyhan, M Sylvester, M Hsieh, A Tathireddy, Y-J Hsieh, AC Matin. 2014. Sigma S-dependent antioxidant defense protects stationary phase Escherichia coli against the bactericidal antibiotic gentamicin. Antimicrob. Agents Chemother. 58(10): 5964-5975 2. Boone T, Cohen A, Chin M, Frederick C, Jackson E, Keyhan M, Lera M, Matin AC, Mayer D, Christopher M, Parra M, Ricco A, Spremo S. 2014. E. coli antimicrobial satellite (EcAMSat): Science payload system development and test. SSC14-WK-8 3. A.C. Matin, A.J. Ricco, M. Parra, K. Ronzano, M. Lera, D. Wu, M. Tan, S. Ross, M. Keyhan, C. Mayberry, C. Beasley, and S. Choi EcAMSat: Nanosatellite Microfluidic Payload Determines Spaceflight Effects on Bacterial Antibiotic Resistance and Their Genetic Basis (2012), 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, 11/28-12/2/12, abstr TVI-6. 2. 4. M. Lera, M. Parra, D. Wu, M. Tan, A. J. Ricco, K. Ronzano, S. Ross, C. Mayberry, C. Beasley, S. Choi, A.C. Matin, and M. Keyhan. Newly Modified Microfluidics Hardware with Flight Heritage Provides Platform for Conducting Inexpensive On-Orbit Microbiology Experiments (2012), 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, 11/28-12/2/12, abstr IPS-14. 3. 5. A.C. Matin, A.J. Ricco, M. Parra, K. Ronzano, M. Lera, D. Wu, M. Tan, S. Ross, M. Keyhan, C. Mayberry, C. Beasley, S. Choi, M. Benoit, J. Wang, R. Singh. Nanosatellite microfluidic payload will determine spaceflight effects on bacterial antibiotic resistance and its genetic basis. ELGRA News, vol 28, September 2013. 6. Tori Chinn, Travis Boone, Matthew Chin, Aaron Cohen, Charlie Friedericks, Evan Jackson, Matthew Lera, AC Matin, Christopher Middour, Macarena Parra, Antonio Ricco, Stevan Spremo. EcAMSat Fluidic System: Lessons Learned for Reduced Gravity Biological Research. ASGSR; September 2014; Pasadena, CA. Session 10, October 25, 2014 7. Aaron Cohen, Matthew Chin, Tori Chinn, Greg Defouw, Charlie Friedericks, Ariel Gatti, Michael Henschke, Mahmonir Keyhan, Matthew Lera, AC Matin, Michael Padgen, Macarena Parra, Antonio Ricco, Stevan Spremo, Dave Squires, Sasha Weston. 2014. E. coli AntiMicrobial Satellite (EcAMSat): NASA's First 6U CubeSat. Submitted.
Bibliography Type:	Description: (Last Updated: 03/27/2020)
Abstracts for Journals and Proceedings	Chinn T, Boone T, Chin M, Cohen A, Friedericks C, Jackson E, Lera M, Matin AC, Middour C, Parra M, Ricco A, Spremo S. "EcAMSat Fluidic System: Lessons Learned for Reduced Gravity Biological Research." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. Session 10, October 25, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Articles in Peer-reviewed Journals	Wang JH, Singh R, Benoit M, Keyhan M, Sylvester M, Hsieh M, Thathireddy A, Hsieh YJ, Matin AC. "Sigma S-dependent antioxidant defense protects stationary-phase Escherichia coli against the bactericidal antibiotic gentamicin." Antimicrob Agents Chemother. 2014 Oct;58(10):5964-75. http://dx.doi.org/10.1128/AAC.03683-14 ; Epub 2014 Jul 28. PubMed PMID: 25070093; PubMed Central PMCID: PMC4187989 , Oct-2014